

**Topic 5**

# Mensuration

**Concept guide · Worked examples · Cambridge-style practice · Fully explained answers**

**Learning focus**

Explain every method before practising. Use the original practice questions to check calculation, reasoning and communication.

**Topic map**

|            |                                       |
|------------|---------------------------------------|
| <b>5.1</b> | Units and compound measures           |
| <b>5.2</b> | Perimeter and area                    |
| <b>5.3</b> | Circles, sectors and arcs             |
| <b>5.4</b> | Volumes and surface areas             |
| <b>5.5</b> | Similar solids and practical problems |

Core foundations and Extended applications are integrated throughout. Study each example, then attempt the practice paper without the notes.

# 1. Units and area

Convert lengths before calculating. Area is measured in square units:  $1 \text{ m}^2 = 10000 \text{ cm}^2$ ; volume is in cubic units:  $1 \text{ m}^3 = 1000000 \text{ cm}^3$ . A triangle has area  $\frac{1}{2} \times \text{base} \times \text{perpendicular height}$ , and a trapezium has area  $\frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$ .

## Worked example

For a trapezium with parallel sides 7 cm and 11 cm and height 4 cm, area =  $(7 + 11) \times \frac{4}{2} = 36 \text{ cm}^2$ .

Exam check: The sloping edge of a triangle is not its perpendicular height unless it forms a right angle to the base.

# 2. Circles, arcs and sectors

The circle circumference is  $2\pi r$  and area is  $\pi r^2$ . A sector with central angle  $\theta$  takes fraction  $\frac{\theta}{360}$  of a full circle, so its arc length is  $(\frac{\theta}{360})2\pi r$  and area is  $(\frac{\theta}{360})\pi r^2$ . Include straight radii when asked for sector perimeter.

## Worked example

Radius 6 cm, angle  $60^\circ$ : arc =  $(\frac{60}{360})2\pi(6) = 2\pi \text{ cm}$ ; sector area =  $(\frac{60}{360})\pi(6^2) = 6\pi \text{ cm}^2$ .

Exam check: Distinguish the curved arc from the full perimeter of a sector.

# 3. Prisms and cylinders

For a prism, volume = cross-sectional area  $\times$  length. For a cylinder, volume =  $\pi r^2 h$  and curved surface area =  $2\pi r h$ . Total closed surface area includes both circular ends; distinguish an open container from a closed one.

## Worked example

A cylinder with  $r = 3 \text{ cm}$  and  $h = 8 \text{ cm}$  has volume  $\pi(3^2)(8) = 72\pi \text{ cm}^3$ .

Exam check: A diameter of 6 cm means  $r = 3 \text{ cm}$ ; do not use 6 in  $\pi r^2$ .

# 4. Cones, spheres and pyramids

A cone has volume  $(\frac{1}{3})\pi r^2 h$  and curved surface area  $\pi r l$ , where  $l$  is slant height. A sphere has volume  $(\frac{4}{3})\pi r^3$  and surface area  $4\pi r^2$ . A pyramid has volume  $(\frac{1}{3}) \times \text{base area} \times \text{perpendicular height}$ . Pythagoras often connects a cone height, radius and slant height.

## Worked example

For a cone with  $r = 3 \text{ cm}$  and  $h = 4 \text{ cm}$ ,  $l = 5 \text{ cm}$ , so curved area =  $15\pi \text{ cm}^2$  and volume =  $12\pi \text{ cm}^3$ .

Exam check: Use the perpendicular height for volume and slant height for curved surface area.

## 5. Composite and similar solids

Break a compound solid into standard parts. Add volumes of joined solids; subtract removed hollows. If lengths are multiplied by  $k$ , areas multiply by  $k^2$  and volumes by  $k^3$ . For material cost, use the actual exposed surface, including or excluding joined faces as appropriate.

### Worked example

Two similar solids have length ratio 2:5. Their volume ratio is  $2^3:5^3 = 8:125$ .

Exam check: For a joined solid, an internal contact face is not part of the external surface.

## Before the practice paper

### Method and accuracy

Write the formula or first operation, show the intermediate step, give units when needed and check by substitution, estimation or a sketch. On non-calculator work, keep exact forms when requested.



# Practice paper

## Topic 5 · Mensuration

### Instructions

Original Cambridge-style practice. Answer all questions. Section A is multiple choice; Section B requires working and final answers. Try Section A without a calculator; use a calculator in Section B only where needed.

|                 |  |
|-----------------|--|
| Candidate name: |  |
| Class / date:   |  |

**Total: 32 marks**

## Section A · Multiple choice

Choose one answer for each question.

1. 1 m<sup>2</sup> equals how many cm<sup>2</sup>? [1]

- A 100
- B 1000
- C 10000
- D 1000000

2. Area of a triangle with base 8 cm and height 5 cm? [1]

- A 13
- B 20
- C 40
- D 80

3. Circumference of a circle with radius 4? [1]

- A  $4\pi$
- B  $8\pi$
- C  $16\pi$
- D  $32\pi$

4. Volume of a cylinder with radius 2 and height 5? [1]

- A  $10\pi$
- B  $20\pi$
- C  $25\pi$
- D  $40\pi$

5. Area of a 90° sector with radius 4? [1]

- A  $2\pi$
- B  $4\pi$
- C  $8\pi$
- D  $16\pi$

6. Volume of a cone with radius 3 and height 6? [1]

- A  $6\pi$
- B  $18\pi$
- C  $27\pi$
- D  $54\pi$

7. Surface area of a sphere with radius 2? [1]

- A  $8\pi$
- B  $12\pi$
- C  $16\pi$
- D  $32\pi$

8. Length scale factor 4 gives volume factor [1]

- A 4
- B 8
- C 12
- D 64

## Section B · Structured questions

9. A sector has radius 8 cm and central angle  $90^\circ$ .

(a) Find its arc length in terms of  $\pi$ . [2]

(b) Find its area in terms of  $\pi$ . [2]

(c) Find its perimeter in terms of  $\pi$ . [2]

**[Total: 6]**

10. A closed cylinder has radius 3 cm and height 10 cm.

(a) Calculate its volume in terms of  $\pi$ . [2]

(b) Calculate its curved surface area in terms of  $\pi$ . [2]

(c) Calculate its total surface area in terms of  $\pi$ . [2]

**[Total: 6]**

11. A cone has radius 5 cm and perpendicular height 12 cm.

(a) Find its slant height. [2]

(b) Find its volume in terms of  $\pi$ . [2]

(c) Find its curved surface area in terms of  $\pi$ . [2]

**[Total: 6]**

12. Two similar cylinders have radii in the ratio 3:5. The smaller has volume  $54 \text{ cm}^3$  and surface area  $72 \text{ cm}^2$ .

(a) Find their volume ratio. [2]

(b) Find the larger volume.

[2]

(c) Find the larger surface area.

[2]

**[Total: 6]**



# Worked solutions

**Fold back this section while completing the practice paper.**

## Marking routine

Award method marks when the mathematical setup is correct even if an arithmetic slip follows.  
Check each part independently.

## Section A · Answers and reasons

1. C — 10000.  $1\text{ m} = 100\text{ cm}$ ; square the factor to obtain 10000.
2. B — 20. Triangle area is  $\frac{1}{2} \times \text{base} \times \text{perpendicular height}$ .
3. B —  $8\pi$ . Circumference  $= 2\pi r = 8\pi$ .
4. B —  $20\pi$ . Volume  $= \pi r^2 h = \pi(2^2)(5) = 20\pi$ .
5. B —  $4\pi$ . Quarter-circle area  $= \pi(4^2)/4 = 4\pi$ .
6. B —  $18\pi$ . Cone volume  $= \pi r^2 h/3 = \pi(9)(6)/3$ .
7. C —  $16\pi$ . Sphere area  $= 4\pi r^2 = 4\pi(4)$ .
8. D — 64. Volume factor is the cube of the length factor:  $4^3$ .

## Section B · Worked solutions

9. a)  $(90/360) \times 2\pi(8) = 4\pi \text{ cm}$  [2]. b)  $(90/360) \times \pi(8^2) = 16\pi \text{ cm}^2$  [2]. c)  $4\pi + 8 + 8 = 16 + 4\pi \text{ cm}$  [2].

10. a)  $\pi(3^2)(10) = 90\pi \text{ cm}^3$  [2]. b)  $2\pi(3)(10) = 60\pi \text{ cm}^2$  [2]. c)  $60\pi + 2\pi(3^2) = 78\pi \text{ cm}^2$  [2].

11. a)  $l = \sqrt{5^2 + 12^2} = 13 \text{ cm}$  [2]. b)  $(1/3)\pi(5^2)(12) = 100\pi \text{ cm}^3$  [2]. c)  $\pi(5)(13) = 65\pi \text{ cm}^2$  [2].

12. a)  $3^3:5^3 = 27:125$  [2]. b)  $54 \times 125/27 = 250 \text{ cm}^3$  [2]. c)  $72 \times 25/9 = 200 \text{ cm}^2$  [2].

### Review your errors

Mark the first wrong step: setup, algebra, arithmetic, units, graph or reasoning. Redo that part without seeing the solution.

Syllabus reference: Cambridge IGCSE Mathematics 0580, 2025-2027,  
[www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf](https://www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf). Paper style informed by the June 2026 papers and  
examiner report provided by the user. All practice questions are original.